

So you want to be a 2n-aire

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Codes

AI Code Tips

Description

So you want to be a 2ⁿ-aire

The player starts with a prize of \$1, and is asked a sequence of n questions. For each question, he may

- quit and keep his prize.
- answer the question. If wrong, he quits with nothing. If correct, the prize is doubled, and he continues with the next question.

After the last question, he quits with his prize. The player wants to maximize his expected prize.

Once each question is asked, the player is able to assess the probability p that he will be able to answer it. For each question, we assume that p is a random variable uniformly distributed over the range $t \dots 1$.

Input is a number of lines, each with two numbers: an integer $1 \leq n \leq 30$, and a real $0 \leq t \leq 1$. Input is terminated by a line containing 0 0. This line should not be processed.

For each input n and t , print the player's expected prize, if he plays the best strategy. Output should be rounded to three fractional digits.

Input

Please Input Input Here

Output

Please Input Output Here

Samples

input	Copy
1 0.5 1 0.3 2 0.6 24 0.25 0 0	

output

Copy

```
1.500
1.357
2.560
230.138
```

Submit

Codes

[HZNUOJ](#) is based on [HUSTOJ](#)

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Server Time: 2025-8-26 15:07:45